Evaluation pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Evaluation pipelines before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Integration tests and fixtures

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Testing pure components

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Configuration and secrets

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Evaluation pipelines preserve case versions, procedure versions and **raw outcome**s before producing metrics. A rerun must identify what changed.

Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

Practise with fading help

> Complete the supported Evaluation pipelines trace and record the first step you can explain without help.

Use W2: Trace the worked case for Evaluation pipelines. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Evaluation pipelines explanation against the supplied facts before discussing it.

Use the worked case for Evaluation pipelines. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Evaluation pipelines facts and keep your

> reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Evaluation pipelines facts and keep your reasoning.

Six cases are planned; five run, four pass and one fails. One is skipped. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Evaluation pipelines facts and keep your reasoning.

A report omits this qualification: Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Evaluation pipelines facts and keep your reasoning.

Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Evaluation pipelines facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Evaluation pipelines to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled evaluation pipelines **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of evaluation pipelines.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

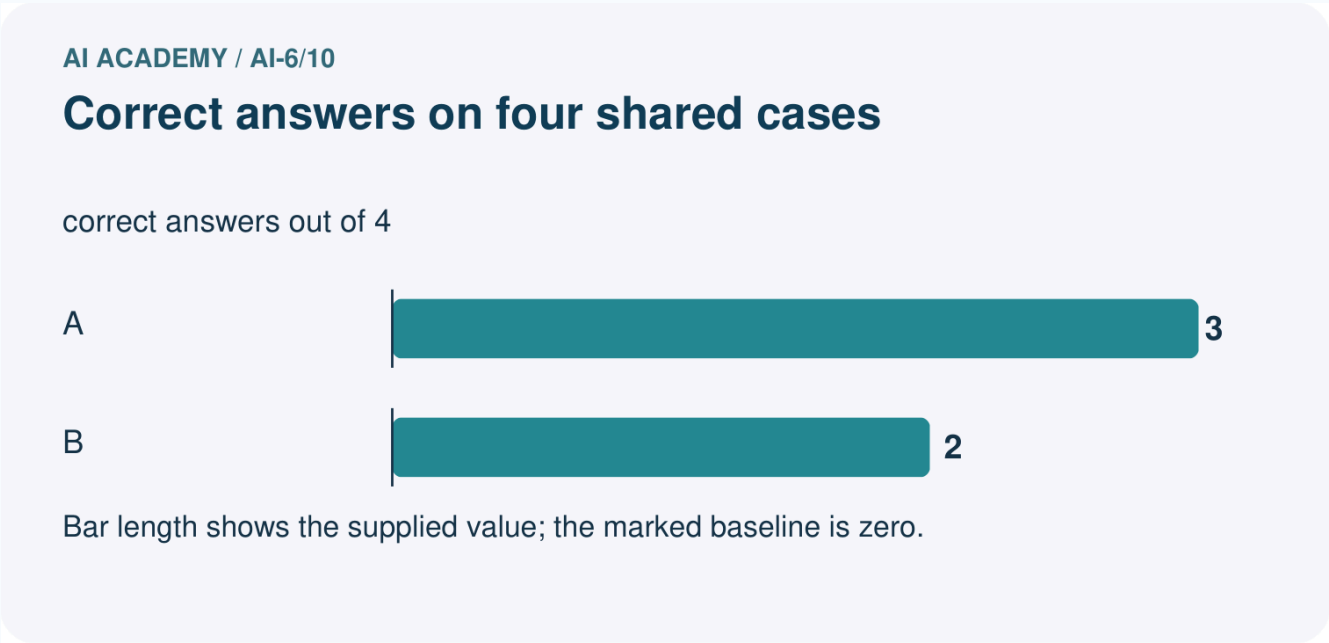
WORKBOOK · 12 OF 15

Pictorial case practice: Evaluation pipelines

> Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels. Explain the labelled step before checking the changed condition.

Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.

A scores [1,1,0,1]; B scores [1,0,0,1]. Each 1 means agreement with the reference answer.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/10

Evaluation pipelines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze cases

Keep questions and reference labels unchanged for both versions.



2. Compute rates

A is 3/4=75%; B is 2/4=50%.



3. Inspect failures

Case two separates the versions; case three fails for both.

CHANGE ONE THING

Remove the shared failed third case after scoring.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Case two separates the versions; case three fails for both.

1. Freeze cases

2. Inspect failures

3. Compute rates

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect failures Case two separates the versions; case three fails for both. A pipeline preserves comparable cases, version labels and scoring.

Report the denominator and disagreements alongside percentages.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove the shared failed third case after scoring. Which conclusion is supported by the supplied facts?

1. Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. The population changed after seeing failures, concealing a weakness.

Repair a missing building block

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

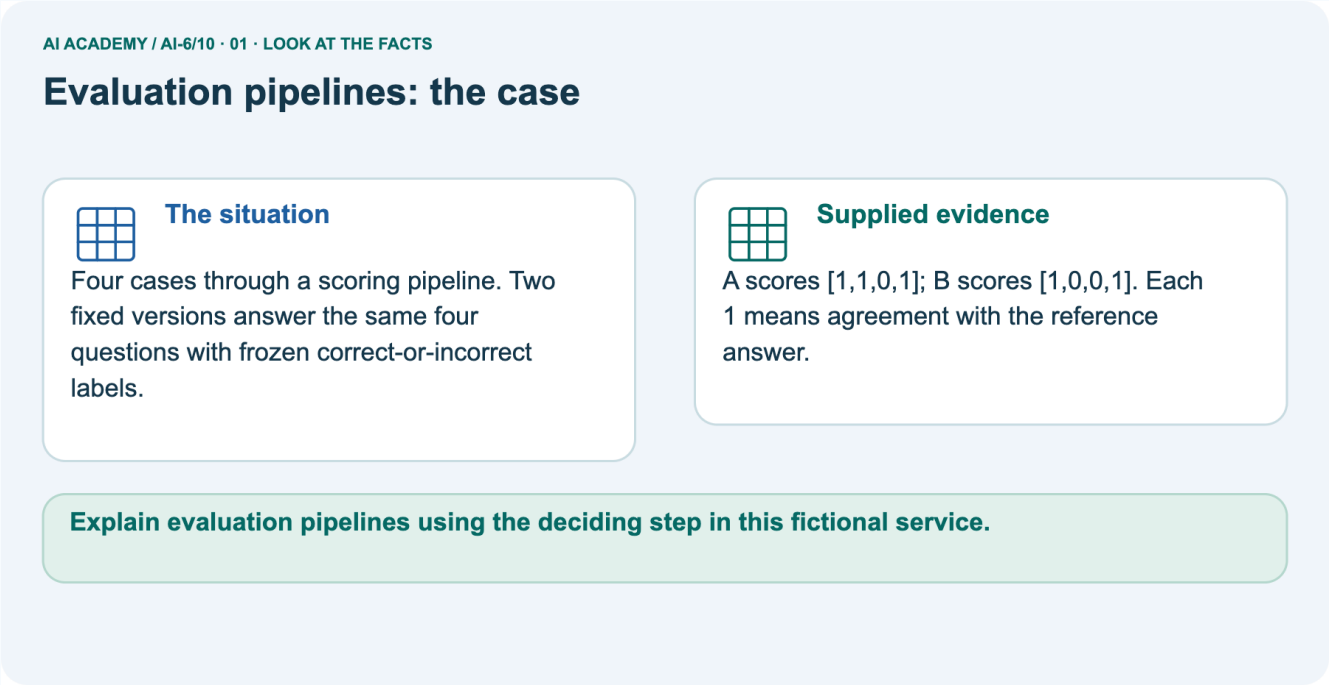
> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Evaluation pipelines picture from its labels, then record a relevant limit.

01. Evaluation pipelines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

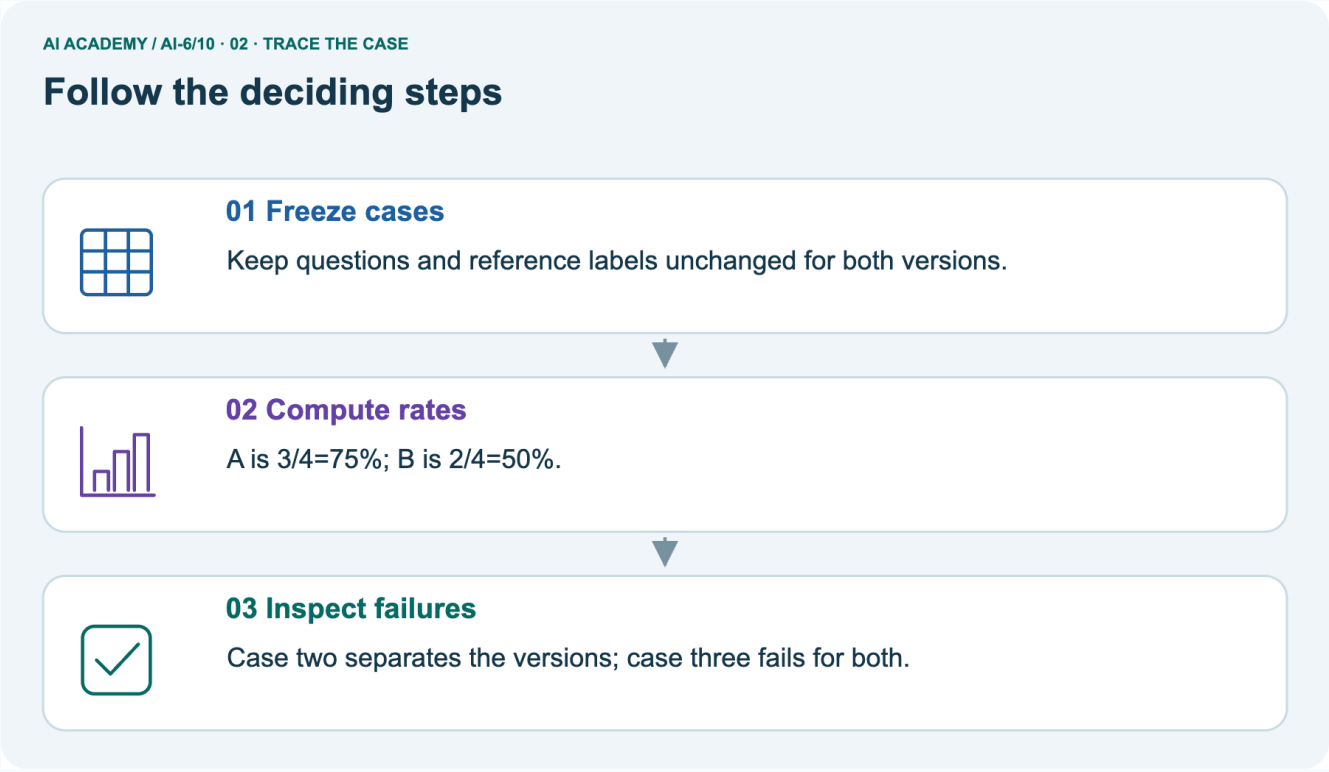
Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels. Supplied facts: A scores [1,1,0,1]; B scores [1,0,0,1].

Each 1 means agreement with the reference answer.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



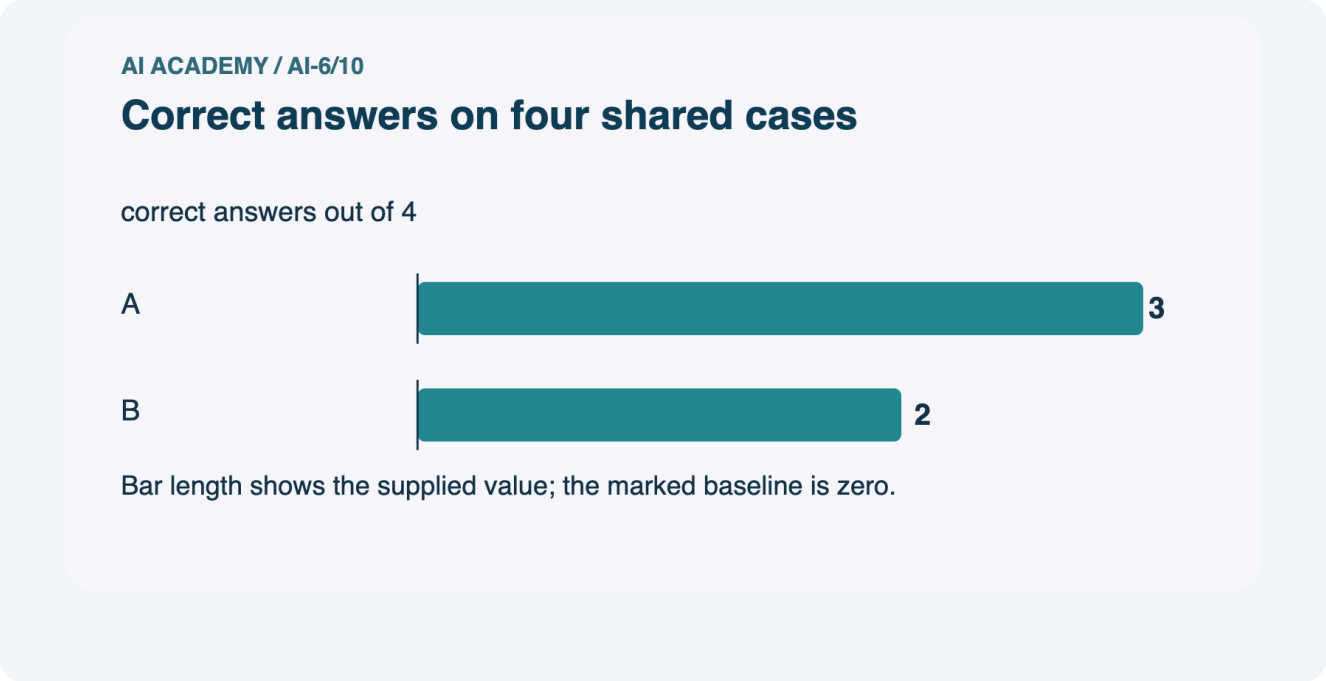
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze cases: Keep questions and reference labels unchanged for both versions. → Compute rates: A is 3/4=75%; B is 2/4=50%. → Inspect failures: Case two separates the versions; case three fails for both.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Correct answers on four shared cases



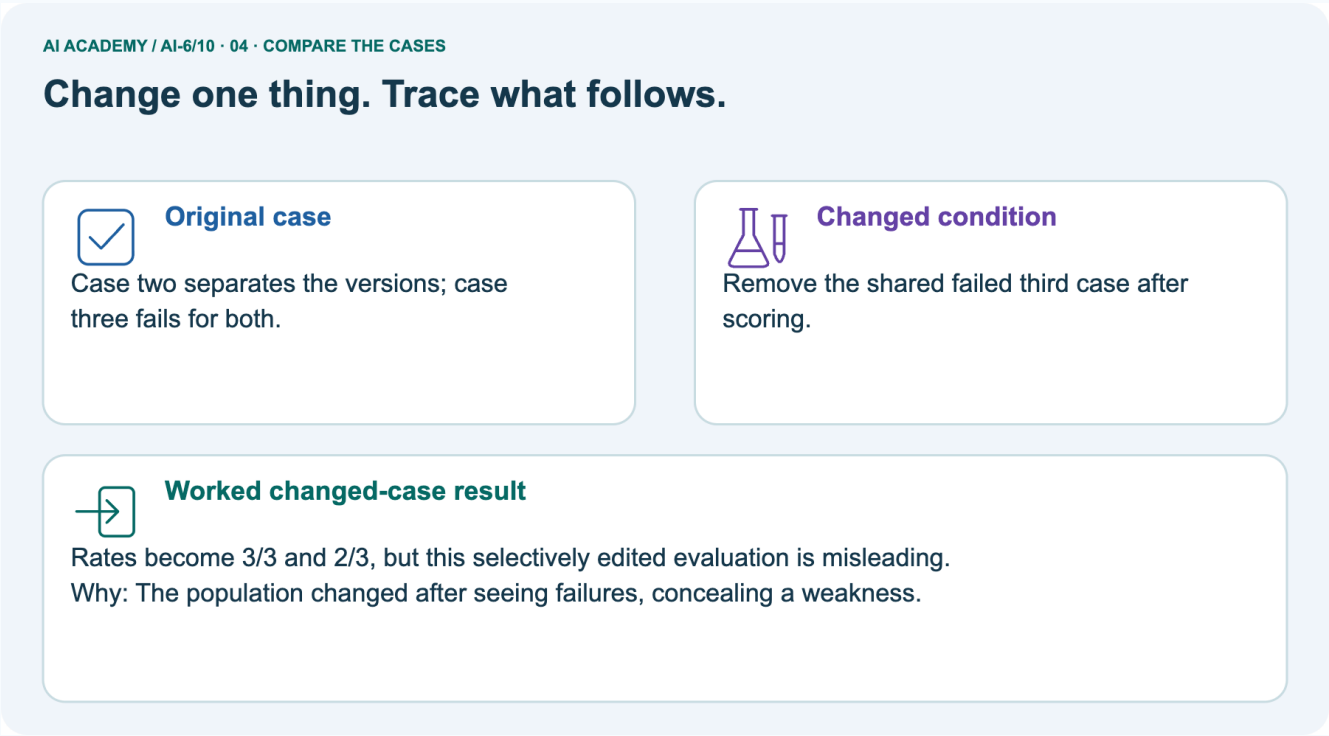
Follow the labelled picture with the matching explanation. Open the vector diagram

Correct answers on four shared cases. Units: correct answers out of 4. A: 3. B: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

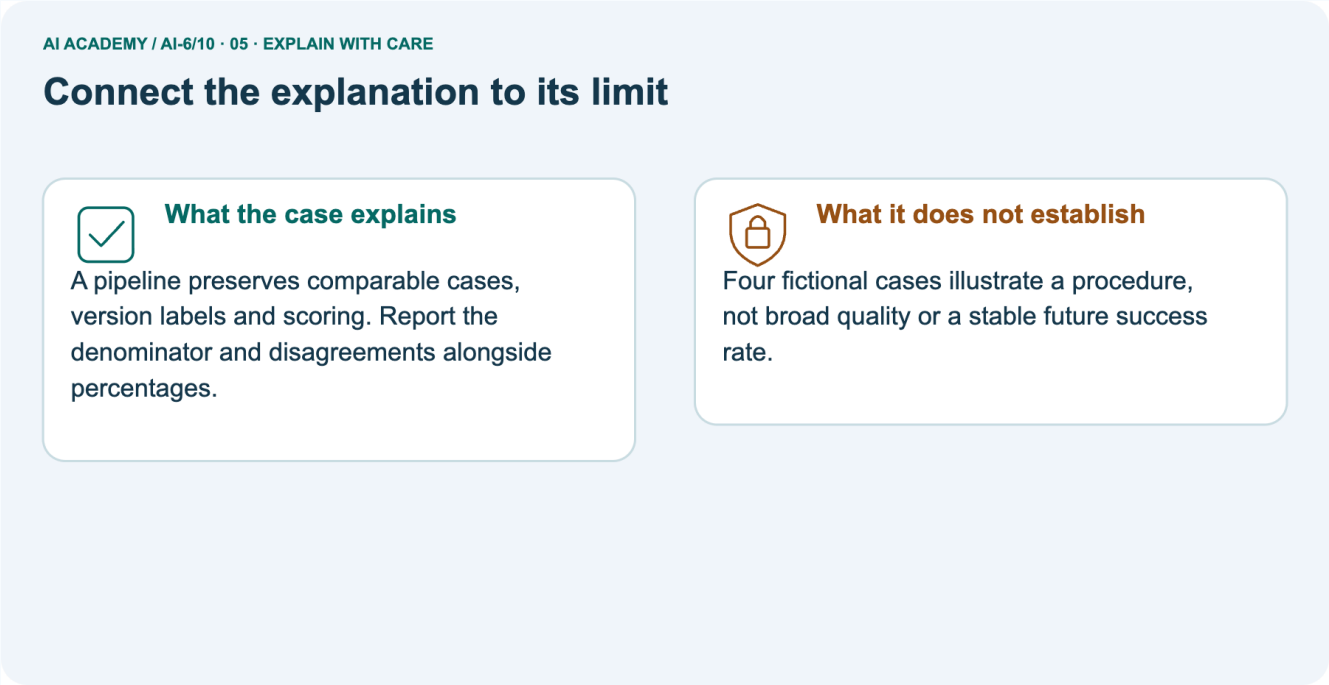
Original case: Case two separates the versions; case three fails for both. Changed condition: Remove the shared failed third case after scoring.

Expected result: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. Reason: The population changed after seeing failures, concealing a weakness.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

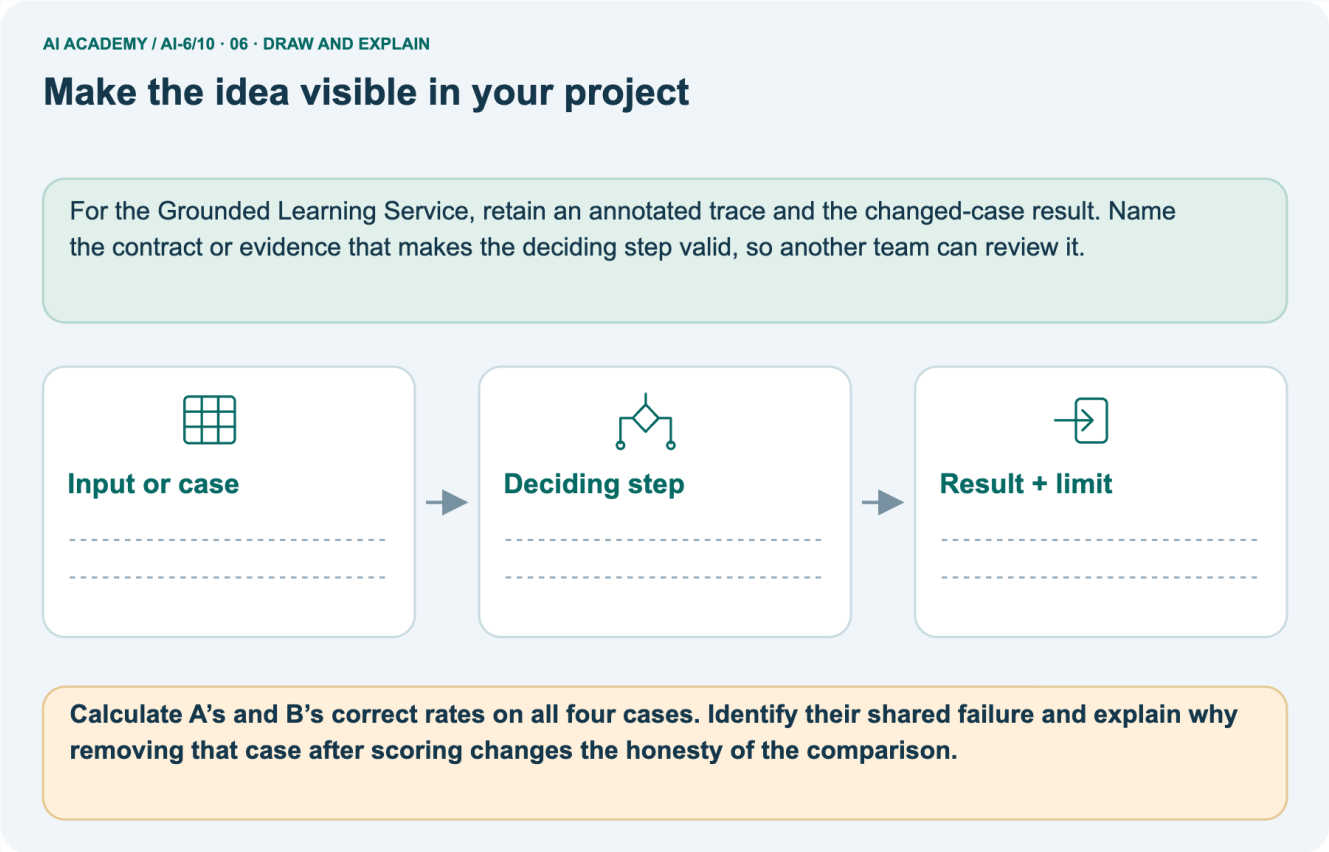
A pipeline preserves comparable cases, version labels and scoring. Report the denominator and disagreements alongside percentages.

Limit: Four fictional cases illustrate a procedure, not broad quality or a stable future success rate.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.